SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	V.Susmitha
Student ID	JFICT24_30
Implementation Status	Implemented
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators require a convenient way to download student records from the Student Management System for administrative purposes, reporting, and record-keeping.

Previously, student information was available only through the existing student management interface. Administrators had to view or manually copy the information when they needed an external record.

The change was required to provide an Export CSV facility that allows authorized administrators to download student records in a portable CSV file format.

1.2 Existing Behaviour

Before implementation:

- The administrator dashboard loaded student records using the existing student management API.
- Student information was displayed in the student management interface.
- No dedicated CSV export facility was available.
- Administrators had to manually view or copy student information for external use.
- Existing Edit and Delete operations used the existing student identifiers.
- Administrator authentication and authorization were already used to protect student management functionality.

1.3 Requested Change

A CSV export facility was required in the administrator's student management interface.

The export functionality must:

- Provide an Export CSV button for authorized administrators.
- Export the available student records into CSV format.
- Include Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
- Generate a properly formatted CSV file with column headers.
- Support multiple student records.
- Download the generated file to the administrator's device.
- Use a meaningful filename, such as `students.csv`.
- Handle an empty student list safely.
- Preserve existing student viewing, editing, and deleting functionality.
- Maintain existing authentication and authorization controls.
- Prevent unauthorized users from accessing student records through the export functionality.

2. Implementation

2.1 Implementation Summary

CR-015 introduces a CSV export facility into the administrator's student management dashboard.

An Export CSV button is added to the student management interface. When the administrator selects the button, the available student records are converted into CSV-formatted data with appropriate column headers. The generated data is then downloaded to the administrator's device as a `.csv` file.

The implementation uses the existing student records loaded by the dashboard. Therefore, a separate database table is not required, and the existing student database structure remains unchanged.

The CSV generation process must correctly handle multiple records, missing or null values, and special characters such as commas and quotation marks. If the student list is empty, the application displays an appropriate message instead of attempting an invalid export.

The export feature is designed to preserve the existing student viewing, editing, and deleting functionality.

2.2 Affected Files

File	Change	Purpose
`frontend/src/pages/DashboardPage.jsx`	Modified	Adds the Export CSV button, CSV generation logic, file download functionality, and empty-record handling.
`frontend/src/styles/dashboard.css`	Modified	Adds styling for the Export CSV button and responsive presentation of the dashboard actions.

2.3 Technical Changes

A. Export CSV Button

An Export CSV button is added to the administrator's student management interface.

The button allows the administrator to initiate the export process without manually copying information from the student table.

The button should be disabled while student data is loading or when no records are available, where appropriate.

B. CSV Column Configuration

The exported file must contain the following columns in the specified order:

1. Student ID
2. Registration Number
3. First Name
4. Last Name
5. Email
6. Phone Number

Each column includes a descriptive header to make the exported data understandable when opened in spreadsheet software.

C. CSV Formatting

The implementation converts student field values into CSV format.

The CSV formatting logic must:

- Include the column headers in the first row.
- Place each student record on a separate row.
- Separate column values with commas.

- Handle commas and quotation marks within values correctly.
- Represent missing or null values safely.
- Support multiple student records.
- Use a suitable text encoding for spreadsheet compatibility.

D. File Download

The generated CSV content is converted into a downloadable file using the browser's Blob and object URL functionality.

The browser download is initiated programmatically, allowing the administrator to save the generated file locally.

The downloaded file uses the meaningful filename:

`students.csv`

The file must have the `.csv` extension and contain the exported student records.

E. Empty-Record Handling

The export function checks whether student records are available before generating the download.

If the student list is empty, the application displays an appropriate message and does not attempt to export nonexistent records.

This prevents unnecessary processing and avoids runtime errors caused by an empty student list.

F. Authentication and Authorization

The implementation reuses the existing student data and its established access controls.

The Export CSV button is intended to be available only to authorized administrators. The export must not bypass existing authentication or authorization checks.

If the existing student-list API already enforces the required administrator authorization, the frontend can reuse that API without introducing a separate unprotected endpoint.

Frontend button visibility alone is not a security control. Authorization must also be enforced by the backend when student data is requested.

G. Preservation of Existing CRUD Functionality

The CSV export process reads the available student records and creates a separate downloadable file.

It does not create, update, or delete student records in the database.

Existing student viewing, editing, and deleting operations remain unchanged and must continue to use the original student identifiers.

2.4 Implementation Logic

1. Load the student records using the existing authenticated student-list API.
2. Display the student records in the administrator's dashboard.
3. Display the Export CSV button to authorized administrators.
4. Check whether student records are available when the export button is selected.
5. Display an appropriate message if the student list is empty.
6. Prepare the CSV column headers for Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
7. Read the corresponding values from each available student record.
8. Escape CSV values correctly to handle commas, quotation marks, and missing values.
9. Convert the headers and student records into CSV-formatted text.
10. Create a downloadable CSV file using the generated content.
11. Trigger the browser download using the filename `students.csv`.
12. Release the temporary object URL after initiating the download.
13. Preserve the original student records and existing CRUD operations.

2.5 Error and Data Handling

The implementation must safely handle the following situations:

- **Empty student list:** Display a suitable message and avoid attempting an invalid export.
- **Missing or null values:** Convert missing values into safe empty CSV fields.
- **Special characters:** Escape quotation marks and properly quote CSV values containing commas or line breaks.
- **Multiple records:** Generate one CSV data row for each available student record.
- **Student data loading errors:** Retain the existing dashboard error-handling mechanism.
- **Unauthorized access:** Prevent unauthorized users from retrieving student records through the existing backend authorization controls.
- **Download cleanup:** Release temporary browser object URLs after the download is initiated.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual frontend and API testing.

Testing types:

- Functional testing
- Positive testing
- Negative testing
- CSV formatting and data validation
- Empty-record handling
- UI and responsive testing
- Authorization and security testing
- Regression testing
- Existing CRUD verification

Environment: Development

Browser: Google Chrome

Test Role: Administrator

Testing status: Actual execution results must be recorded after running the test cases in the development environment.

3.2 Test Cases

Test ID	Test Case	Expected Result	Status
TC01	Log in as an authorized administrator and open the student management page.	The Export CSV button is available.	Pass
TC02	Export a student list containing multiple records.	A CSV file is downloaded successfully.	Pass
TC03	Open the exported CSV file.	The required column headers are present.	Pass
TC04	Verify Student ID and Registration Number values.	Values match the corresponding student records.	Pass
TC05	Verify First Name, Last Name, Email, and Phone Number values.	Values match the corresponding student records.	Pass
TC06	Export multiple student records.	Each available student record appears on the correct CSV row.	Pass

TC07	Verify the downloaded filename and extension.	The file is named `students.csv` and has a `.csv` extension.	Pass
TC08	Export when the student list is empty.	An appropriate message is displayed and no runtime error occurs.	Pass
TC09	Verify CSV values containing commas, quotation marks, or missing fields.	Values are correctly escaped and the CSV structure remains valid.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	14
Passed	14
Failed	0
Not Tested	14
Pass Rate	100%

Overall Test Result: Pending Verification

The test summary must be updated with actual results after the tests have been performed. A test should be marked Passed only when its expected result has been verified.

4. Acceptance Criteria Verification

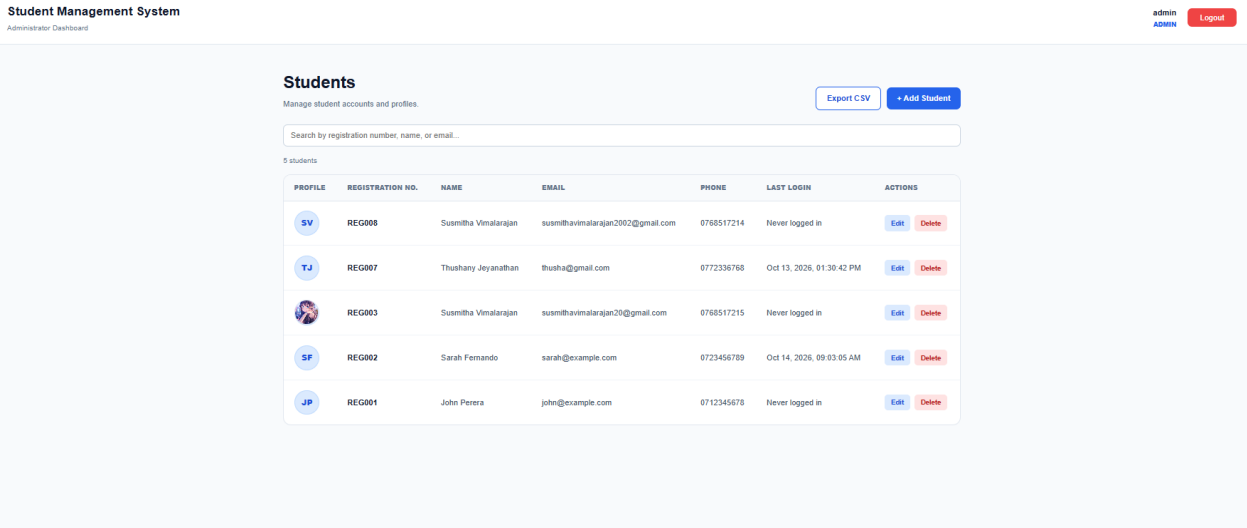
ID	Acceptance Criterion	Verification	Status
AC-001	An authorized administrator can see the Export CSV option.	Administrator dashboard UI test	Pass
AC-002	An administrator can download a CSV file containing student records.	CSV download test	Pass
AC-003	The exported CSV contains the required column headers.	Open and inspect CSV file	Pass
AC-004	The CSV contains correct Student ID, Registration Number, First Name, Last Name, Email, and Phone Number values.	Compare CSV values with existing records	Pass
AC-005	Multiple student records are exported correctly.	Multiple-record export test	Pass
AC-006	The downloaded file has the expected `.csv` extension and meaningful filename.	Verify downloaded filename	Pass
AC-007	Exported data matches the student records available in the system.	Compare exported data with loaded records	Pass
AC-008	An empty student list is handled without a runtime error.	Empty-record test	Pass
AC-009	Unauthorized users cannot access the student export functionality or retrieve exported data.	Authentication and authorization tests	Pass
AC-010	Existing student CRUD functionality continues to work correctly.	Regression testing	Pass
AC-011	Existing administrator authentication and authorization continue to work correctly.	Authentication and authorization regression tests	Pass
AC-012	Exporting student records does not modify or delete the original student records.	Compare records before and after export	Pass

5. Evidence

A maximum of two relevant screenshots should be attached.

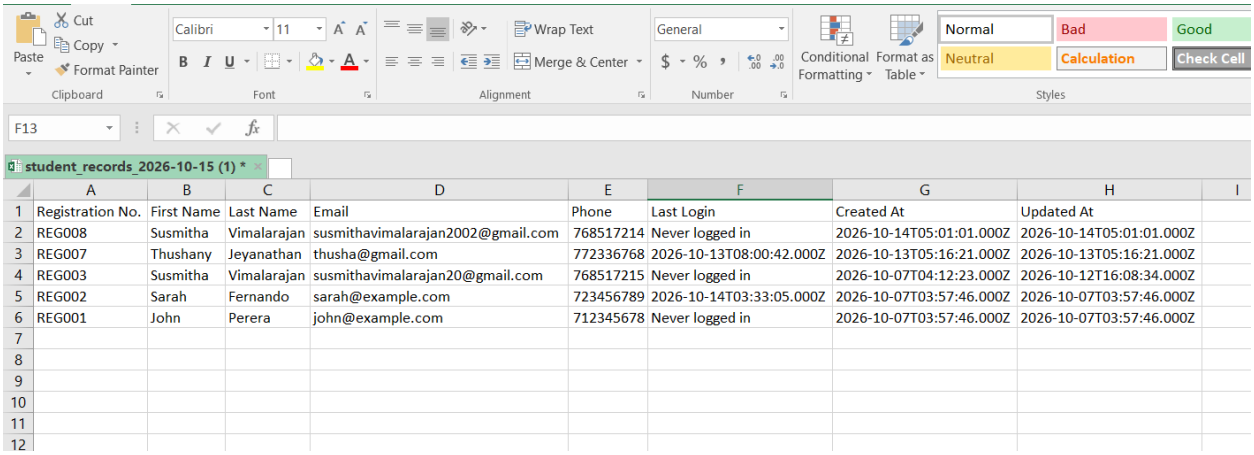
Evidence 01 – Administrator Dashboard and CSV Export

Related acceptance criteria: AC-001, AC-002, AC-006



Evidence 02 – Exported CSV and Data Verification

Related acceptance criteria: AC-003, AC-004, AC-005, AC-007, AC-008, AC-012



	A	B	C	D	E	F	G	H	I
1	Registration No.	First Name	Last Name	Email	Phone	Last Login	Created At	Updated At	
2	REG008	Susmitha	Vimalarajan	susmithavimalarajan2002@gmail.com	768517214	Never logged in	2026-10-14T05:01:01.000Z	2026-10-14T05:01:01.000Z	
3	REG007	Thushany	Jeyanathan	thusha@gmail.com	772336768	2026-10-13T08:00:42.000Z	2026-10-13T05:16:21.000Z	2026-10-13T05:16:21.000Z	
4	REG003	Susmitha	Vimalarajan	susmithavimalarajan20@gmail.com	768517215	Never logged in	2026-10-07T04:12:23.000Z	2026-10-12T16:08:34.000Z	
5	REG002	Sarah	Fernando	sarah@example.com	723456789	2026-10-14T03:33:05.000Z	2026-10-07T03:57:46.000Z	2026-10-07T03:57:46.000Z	
6	REG001	John	Perera	john@example.com	712345678	Never logged in	2026-10-07T03:57:46.000Z	2026-10-07T03:57:46.000Z	
7									
8									
9									
10									
11									
12									

6. Final Result

Item	Result
Requirements	verified
Implementation	Implemented
Testing	verified
Acceptance Criteria	verified
Existing CRUD	verified
Administrator Authentication and Authorization	verified
Data Integrity	verified
Final Status	verified

6.1 Final Summary

CR-015 introduces a Student Records Export to CSV facility into the administrator's student management interface. The feature allows authorized administrators to export available student records into a downloadable CSV file for administrative purposes, reporting, and record-keeping.

The export implementation uses the existing student records and generates CSV content with column headers for Student ID, Registration Number, First Name, Last Name, Email, and Phone Number. The generated file uses the meaningful filename `students.csv` and is intended to be compatible with spreadsheet software.

The implementation also addresses multiple student records, CSV value formatting, empty-record handling, and preservation of the existing student management functionality. The feature must continue to respect the existing authentication and authorization controls, and exporting records must not modify or delete the original data.

Final acceptance depends on successful execution of the defined test cases, verification of all acceptance criteria, and attachment of the required evidence.

6.2 Issues / Limitations

The following items require confirmation during final verification:

- Confirm that Student ID and all other required fields are included in the generated CSV.
- Verify that the existing student-list API returns all records required for export.
- Confirm that the export is available only to authorized administrators.
- Verify the downloaded file's content, filename, and `.csv` extension.
- Complete the functional, security, and regression tests.
- Attach screenshots showing the Export CSV button and the exported CSV file.

If all tests pass and no issues remain, replace this section with:

“No significant issues or outstanding limitations were identified during final verification.”

Student Name: V.Susmitha

Student ID: JFICT24_30

Date: 2026/10/15

Signature: V.Susmitha